Learning From Data

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Contents

1 The Learning Problem **2**
- 1.1 Learning from examples . . . 2
- 1.2 Components of the learning problem . . . 2
- 1.3 The perceptron hypothesis set . . . 3
- 1.4 The perceptron learning algorithm . . . 3
- 1.5 Types of learning . . . 4
- 1.6 A learning puzzle . . . 5

Lecture 1: The Learning Problem

1.1 Learning from examples

Machine learning infers a rule from observations. It is useful when there is structure to exploit, a satisfactory rule is unknown, and examples are available.

Example: *Movie recommendations.* Predict a viewer's rating by matching a vector of viewer preferences with a vector of movie attributes and summing their contributions. Comedy, action, and particular actors suggest possible factors, but the factors themselves can be learned from observed ratings. The data determine the representation as well as the predictions.

1.2 Components of the learning problem

For credit approval, an input describes an applicant, using attributes such as income and debt, and the output indicates whether the applicant is a good credit risk. The ideal decision rule is unknown; historical records provide examples of its behavior.

Definition 1.1 [Learning setup]. Let $\mathcal{X}$ be the input space and $\mathcal{Y}$ the output space. An unknown target function $f : \mathcal{X} \to \mathcal{Y}$ supplies the labels in the training set

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}, \qquad y_n = f(\mathbf{x}_n).$$

A hypothesis set $\mathcal{H}$ contains the candidate functions $h : \mathcal{X} \to \mathcal{Y}$. A learning algorithm $\mathcal{A}$ uses $\mathcal{D}$ to choose a final hypothesis $g \in \mathcal{H}$ that should approximate f on new inputs.

The pair $(\mathcal{H}, \mathcal{A})$ is the learning model. The target and observations describe the problem; the hypothesis set and algorithm are the two components we choose to solve it.

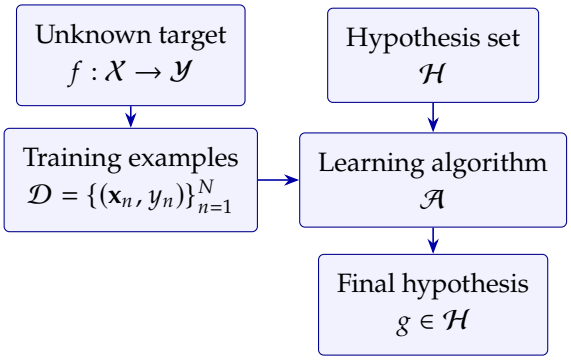


Figure 1: The target supplies training labels. The algorithm sees examples and candidate hypotheses, then returns g; it has no direct access to f.

1.3 The perceptron hypothesis set

For binary credit decisions, take $\mathcal{Y} = \{-1, +1\}$, with $+1$ representing approval. A PERCEPTRON compares a weighted sum of the d input attributes with a threshold τ:

$$h(\mathbf{x}) = \operatorname{sign}\left(\sum_{i=1}^{d} w_i x_i - \tau\right).$$

Absorb the threshold into a bias $w_0 = -\tau$ and append the constant coordinate $x_0 = 1$. From here on, $\mathbf{x} = (1, x_1, \ldots, x_d)^{\mathsf{T}}$ and $\mathbf{w} = (w_0, w_1, \ldots, w_d)^{\mathsf{T}}$, so

$$h(\mathbf{x}) = \operatorname{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x}), \qquad \mathcal{H} = \{\mathbf{x} \mapsto \operatorname{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x}) : \mathbf{w} \in \mathbb{R}^{d+1}\}.$$

In the original attribute space, the decision boundary is $w_0 + \sum_{i=1}^{d} w_i x_i = 0$: a line in two dimensions and a hyperplane in higher dimensions, provided $(w_1, \ldots, w_d) \neq 0$.

Definition: The training data are LINEARLY SEPARABLE if some $\mathbf{w}$ places every example strictly on its correct side: $y_n \mathbf{w}^{\mathsf{T}}\mathbf{x}_n > 0$ for all n.

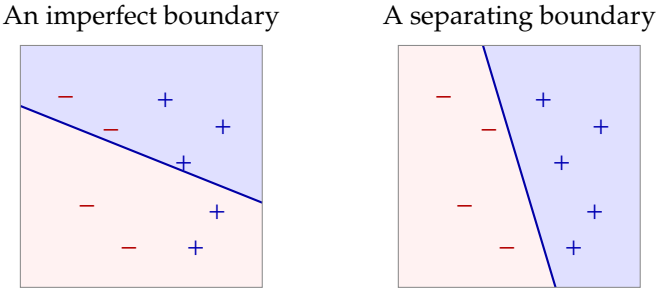


Figure 2: Schematic data with two candidate boundaries. The left boundary makes mistakes; the right one separates all training examples.

1.4 The perceptron learning algorithm

The PERCEPTRON LEARNING ALGORITHM (PLA) repeatedly chooses a misclassified training example $(\mathbf{x}_n, y_n)$ and updates

$$\boxed{\mathbf{w} \leftarrow \mathbf{w} + y_n \mathbf{x}_n.}$$

For the procedure below, a zero score also triggers an update. This avoids assigning a boundary point to either class during training.

Algorithm 1.1: Perceptron learning algorithm

Input: Augmented examples $(\mathbf{x}_n, y_n)$, with $x_{n,0} = 1$ and $y_n \in \{-1, +1\}$
Output: A classifier that separates the training data, if the loop terminates

$\mathbf{w} \leftarrow \mathbf{0}$
while some n satisfies $y_n \mathbf{w}^\mathsf{T} \mathbf{x}_n \le 0$ **do**
    choose such an index n
    $\mathbf{w} \leftarrow \mathbf{w} + y_n \mathbf{x}_n$
return $g(\mathbf{x}) = \operatorname{sign}(\mathbf{w}^\mathsf{T} \mathbf{x})$

For a positive example, add $\mathbf{x}_n$; for a negative one, subtract it. The update moves the chosen example's score toward its correct sign:

$$y_n \mathbf{w}_{\text{new}}^\mathsf{T} \mathbf{x}_n = y_n \mathbf{w}_{\text{old}}^\mathsf{T} \mathbf{x}_n + \|\mathbf{x}_n\|^2.$$

This increases its signed score but need not correct it in one step, and other examples can become misclassified. The augmented update also adjusts the bias: $w_0 \leftarrow w_0 + y_n$.

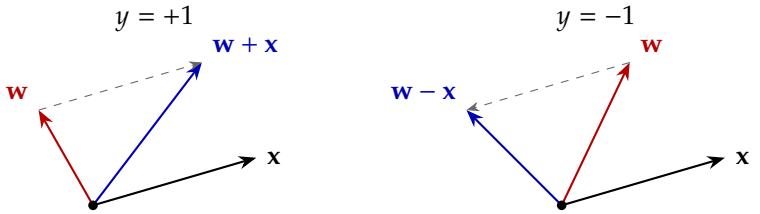


Figure 3: Schematic vector updates for misclassified examples. Dashed arrows complete the addition of $y\mathbf{x}$ to $\mathbf{w}$.

If the procedure stops, all training examples are correctly classified. Without a separating hyperplane, this stopping condition cannot be reached. Agreement on the training set still leaves the question of performance on new inputs.

1.5 Types of learning

The available feedback distinguishes three learning paradigms.

- Supervised learning receives input–label pairs $(\mathbf{x}, y)$. In coin recognition, mass and size are inputs, and the denomination is the label. Labeled examples determine decision regions for future coins.
- Unsupervised learning receives inputs without labels. The same coin measurements can reveal clusters, but their denominations are not supplied.
- Reinforcement learning receives feedback evaluating a chosen output or action, rather than the correct action itself. A backgammon player learns from game outcomes; a win or loss evaluates play without specifying the best move at every position.

1.6 A learning puzzle

Six labeled grids constrain an unknown binary rule. What label should be assigned to the new grid?

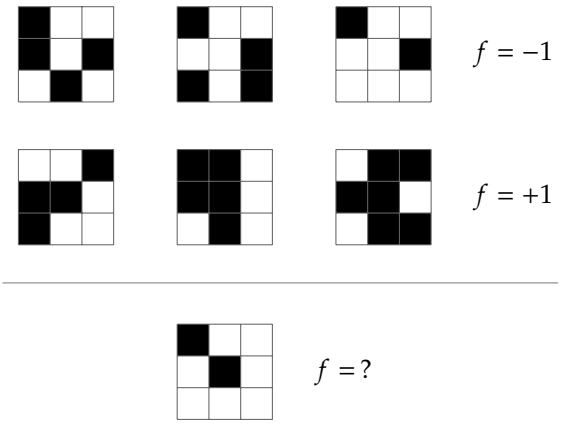


Figure 4: The six examples and unlabeled query in the learning puzzle. Black and white cells are the input attributes.

One consistent rule assigns +1 exactly when the center cell is black. It predicts +1 for the query. Another assigns +1 when the center is black *and* at least three cells are black. It fits the same six examples but predicts −1 for the query, which has only two black cells.

Thus, the examples alone do not uniquely determine the unseen label. Choosing a hypothesis set restricts which rules count as plausible; fitting the observations is only part of the learning problem.

Source. Yaser S. Abu-Mostafa, *Learning From Data*, Lecture 1: The Learning Problem, Caltech, April 3, 2012. Lecture slides.